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PART I

Frontal structure and Antarctic Bottom Water flow through the Princess Elizabeth Trough, Antarctica

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Abstract

Hydrographic, current meter and ADCP data collected during two recent cruises in the South Indian Ocean (*RRS Discovery* cruise 200 in February 1993 and *RRS Discovery* cruise 207 in February 1994) are used to investigate the current structure within the Princess Elizabeth Trough (PET), near the Antarctic continent at 85°E, 63–66°S. This gap in topography between the Kerguelen Plateau and the Antarctic continent, with sill depth 3750 m, provides a route for the exchange of Antarctic Bottom Water between the Australian–Antarctic Basin and the Weddell–Enderby Basin. Shears derived from ADCP and hydrographic data are used to deduce the barotropic component of the velocity field, and thus the volume transports of the water masses. Both the Southern Antarctic Circumpolar Current Front (SACCF) and the Southern Boundary of the Antarctic Circumpolar Current (SB) pass through the northern PET (latitudes 63 to 64.5°S) associated with eastward transports. These are deep-reaching fronts with associated bottom velocities of several cm s^{-1}. Antarctic Bottom water (AABW) from the Weddell–Enderby Basin is transported eastwards in the jets associated with these fronts. The transport of water with potential temperatures less than 0°C is 3 ($\pm$ 1) Sv. The SB is shown to meander in the PET, caused by the cyclonic gyre immediately west of the PET in Prydz Bay. The AABW therefore also meanders before continuing eastwards. In the southern PET (latitudes 64.5 to 66°S) a bottom intensified flow of AABW is observed flowing west. This AABW has most likely formed not far from the PET, along the Antarctic continental shelf and slope to the east. Current meters show that speeds in this flow have an annual scalar mean of 10 cm s^{-1}. The transport of water with potential temperatures less than 0°C is 20 ($\pm$ 3) Sv. The southern PET features westward flow throughout the water column, since the shallower depths

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are dominated by the flow associated with the Antarctic Slope Front. Including the westward flow of bottom water, the total westward transport of the whole water column in the southern PET is 45 (± 6) Sv. © 1999 Elsevier Science Ltd. All rights reserved.

1. Introduction

The physical oceanography of the majority of the region surrounding Antarctica south of 60°S is poorly understood, principally because of its remoteness and winter sea-ice cover. In this paper we present observations taken during 1993 and 1994 in the Princess Elizabeth Trough (PET) in the Indian sector of the Southern Ocean (Fig. 1). The PET is a gap in the topography between the Kerguelen Plateau and the Antarctic continent (85°E, 63–66°S). The sill depth is 3750 m.

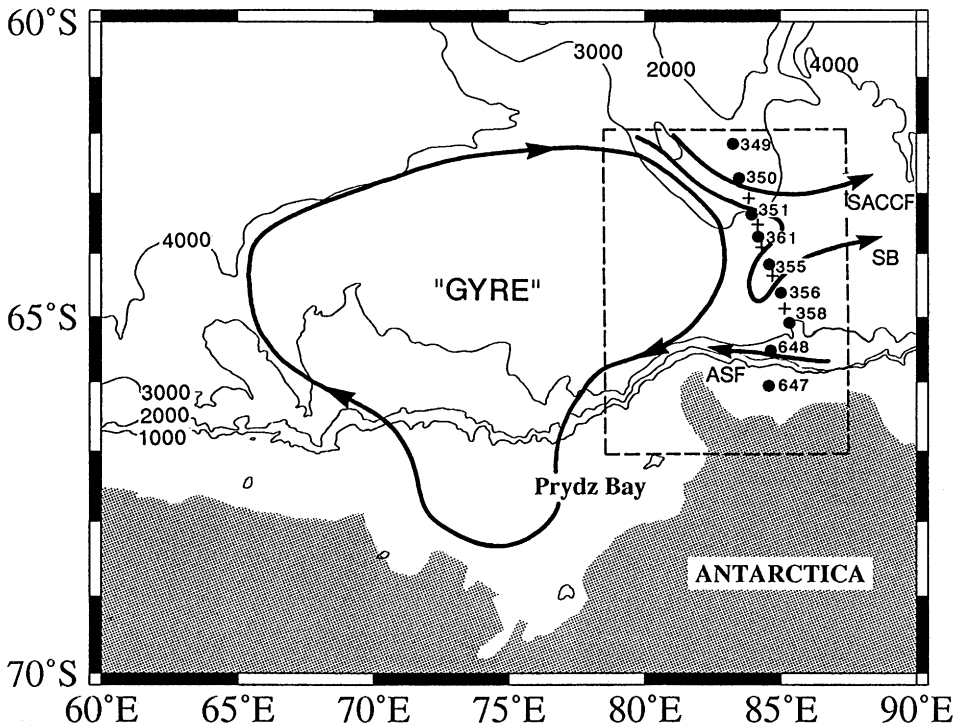

Fig. 1. Schematic of frontal positions in the Princess Elizabeth Trough, showing the Southern ACC Front (SACCF), the Southern Boundary of the ACC (SB) and the Antarctic Slope Front (ASF). Also shown are the locations of the CTD stations (dots) and moorings (+ symbols) carried out during *RRS Discovery* cruises 200 and 207. The cartoon of the Prydz Bay gyre is derived from the analysis of Smith et al. (1984), supplemented by the observations of Nunes Vaz and Lennon (1996) and the results presented in this paper. Bathymetric depth contours are every 1000 m. The dashed box shows the area covered by Fig. 7.

Near the Antarctic continent, in the subpolar regime, westward winds predominate. At the boundary between these winds and the eastward winds north of 60°S is the Antarctic Divergence (AD), which is thought to form a narrow broken circumpolar belt (Deacon, 1937). The divergence in the surface layers causes upwelling of warm, saline Circumpolar Deep Water. Zverev (1963) examined a variety of surface data from early Southern Ocean cruises and suggested that between 20°E and 100°E the circulation consisted of a series of large cyclonic gyres centred on the divergence zone, contrary to the earlier concept of a continuous westward wind drift. Smith et al. (1984), on the basis of sparse hydrographic data, postulated the presence of a large cyclonic gyre centred at about 66°S extending between 73°E and 85°E in the Prydz Bay area. Fig. 1 indicates an approximate location for such a gyre. Wakatsuchi et al. (1994) identified three other gyres centred at approximately 68°E, 100°E and 115°E, showing them to be semi-permanent features, their positions determined by ridges extending northward from the continental shelf. Further support for a gyre in the region of Prydz Bay was provided recently by Nunes Vaz and Lennon (1996).

South of the AD, the Antarctic Slope Front (ASF) is found above the continental slope associated with a westward flowing frontal jet. Jacobs (1991) examined data from the Weddell and Ross Gyres and suggested that the ASF is primarily a subsurface front, surface properties being variable as a result of local melting and freezing or regional convergence or divergence. In the Weddell gyre, Fahrbach et al. (1992) explained that the ASF separates Eastern Shelf Water from Antarctic Surface Water in the surface layers and Circumpolar Deep Water in the deeper layers. The ASF has frequently been shown to have a double-banded structure, although the reasons for this split are not yet certain (Fahrbach et al., 1994; Heywood et al., 1998).

Eastward winds north of about 60°S drive the eastward flowing ACC. The ACC consists of several bands of high speed flow associated with thermohaline fronts (Orsi et al., 1995), the Subantarctic Front, the Polar Front and the Southern ACC Front (SACCF). The Subantarctic Front and Polar Front are not discussed further in this paper. Orsi et al. (1995) studied the circumpolar characteristics of the SACCF using historical data, suggesting that the southern limit of the SACCF coincides with the potential temperature, θ, being less than 0°C along a θ-min at depths less than 150 m. The northern limit is where θ is greater than 1.8°C along a θ-max at depths greater than 500 m, or where the salinity is greater than 34.73 along a S-max at depths deeper than 800 m. The analysis of Sparrow et al. (1996) confirms these criteria as applicable in the South Indian Ocean. Orsi et al. (1995) noted that the boundary between the ACC and subpolar regime – the Southern Boundary (SB) – is sometimes associated with geostrophic shear, and that the SACCF is usually found very close to this boundary (Orsi et al., 1995; Sparrow et al., 1996). The position of the SB is defined by the southern limit of Upper Circumpolar Deep Water (UCDW) (Orsi et al., 1995). UCDW is characterised by a high nutrient and low oxygen content ($< 200\ \mu mol\,l^{-1}$ at a potential density (σ_0) of 27.35–27.75 (Sievers and Nowlin, 1984)) and is not found south of the ACC. Lower Circumpolar Deep Water (LCDW), however, is dense enough to penetrate south of the ACC into the subpolar regime, having a σ_0 of 27.75–27.83 and an associated salinity maximum of greater than 34.75. Apart from the analyses of historical data and model output (Orsi et al., 1995; Sparrow et al., 1996), no

detailed observations of the SB and SACCF have been published, so their structure and associated frontal jets are poorly understood. We have therefore taken the opportunity afforded by the availability of detailed, high resolution hydrographic, current meter and ADCP data to study the structure of the fronts within the PET region.

The flow paths, formation processes and sources of Antarctic Bottom Water (AABW) have attracted much attention. AABW can be distinguished by very cold temperatures ($< 0°C$), high dissolved oxygen ($> 5\ ml\,l^{-1}$), and low salinities (< 34.73) relative to Circumpolar Deep Water, except in the Ross Sea, where the shelf waters and newly formed AABW are particularly saline. AABW formed in the Ross Sea is more saline than that formed in the Weddell Sea, and therefore the AABW in the Australian–Antarctic Basin is about 0.02 more saline than that in the Weddell–Enderby Basin (Mantyla and Reid, 1995). AABW is also known to form along other portions of the Antarctic shelf, particularly the Adelie Coast to the east of the PET (Mantyla and Reid, 1995). A parallel study of the water masses in the PET carried out by Frew et al. (1995) concluded that the AABW found in the PET is isotopically distinct from that found in the southwestern Weddell Sea or the Ross Sea and therefore has a source closer to the PET. From chlorofluoromethane observations of recently ventilated bottom water, Mantisi et al. (1991) suggested that AABW was formed in the Amery Basin, and that water in the southern PET had originated to the east. Speer and Forbes (1994) carried out a Conductivity/Temperature/Depth (CTD) section in the PET to investigate the throughflow of AABW. Using an intermediate level of no motion, based partly on the direction of iceberg drift, they concluded that the most likely flow direction was westward, predicting a transport of 3.7 Sv of deep and bottom (below 2500 m) waters. In this paper we use CTD data in conjunction with ADCP data to calculate the transport of AABW through the PET.

2. Data and methods

Data in the PET were gathered during two recent cruises in the South Indian Ocean: *RRS Discovery* cruise 200 (D200) in February 1993 (Dickson, 1993) and *RRS Discovery* cruise 207 (D207) in February 1994 (Dickson, 1995). During D200, seven full depth CTD stations (12349–12358) were occupied. On D207 two stations (12647 and 12648) were carried out to the south of the previous section (Fig. 1). The CTD multisampler was equipped with a Neil Brown Mk IIIB with Beckman oxygen sensor. A transmissometer attached to the multisampler measured the turbidity of the water, by determining the percentage transmittance of light (per metre), and accounting for the density of sea water (the potential transmittance). A mixture of Sippican T7 (≈ 750 m depth) and T5 (≈ 1800 m depth) Expendable Bathythermographs (XBTs) were deployed in the PET during D207. Continuous thermosalinograph (TSG) data were also available to examine surface features.

High resolution acoustic Doppler current profiler (153.6 kHz) data were collected on both cruises. The ADCP data were processed to obtain absolute velocities using ship's heading from the Ashtech 3 Dimensional General Positioning System (3DGPS)

(King and Cooper, 1993). Without such accurate heading determinations it would not have been possible to use the ADCP data to provide a reference level for the geostrophic shear. Further details of data processing may be found in Sparrow (1996).

3. Results

3.1. CTD sections

Figs. 2 and 3 show potential temperature and salinity sections and surface temperature and salinity traces from the TSG. The TSG plots show the north to south and the south to north crossings of the PET, with a maximum time difference between sampling on the legs of four days. Fig. 4 shows dissolved oxygen and potential transmittance data for the PET for cruise D200. The salinity maximum of LCDW is apparent at about 1000 m, overlain by the oxygen minimum and temperature maximum of UCDW at 500 m. In the bottom water, the isotherms and isohalines sloping upwards towards the continental slope suggest cold, fresh Antarctic Bottom Water flowing to the west hugging the base of the slope, relative to a shallow reference level. This water is relatively young, since it is high in dissolved oxygen. The marked decrease in the clarity of the water near the sea bed in the southern PET suggests comparatively vigorous currents that are stirring up the sediment.

The dip in the isopleths near 63.5°S (best observed in the potential temperature section of Fig. 2 and the dissolved oxygen section of Fig. 4) is characteristic of an eddy-like or meandering structure. In this case, the meandering of the SB in the PET is observed on both southgoing and northgoing TSG traces in 1993, in the XBT section in the following year (1994; see Section 3.4) and in a CTD section carried out in 1992 by Speer and Forbes (1994) (their Fig. 6), and may be semi-permanent.

The SB corresponds to the southern limit of UCDW, characterised by a distinctive oxygen minimum (Orsi et al., 1995). In Fig. 5, potential density is plotted against oxygen and salinity for each CTD station within the PET. Stations 12349 and 12350 were within the ACC regime, since the oxygen minimum of 180 μmol l^{-1} at $\sigma_0 =$ 27.65–27.68 is characteristic of UCDW. Stations 12358 and 12361 were within the subpolar regime, as the oxygen minimum of approximately 220 μmol l^{-1} at $\sigma_0 = 27.81$ corresponds to LCDW only (the salinity maximum of LCDW being at $\sigma_0 =$ 27.80–27.81). The two stations from D207 (12647 and 12648) are obviously well within the subpolar regime (since no bottle samples were taken to calibrate the oxygen sensor on these stations, the oxygen profiles are offset from the D200 profiles). Whether stations 12351, 12355 and 12356 lie within the ACC is less obvious. Station 12356 has an oxygen minimum at $\sigma_0 = 27.77$, below the 27.75 limit of the UCDW signal, and therefore lay just within the subpolar regime. Station 12355 (in the centre of the eddy or meander like structure) has some UCDW characteristics with an oxygen minimum of 190 μmol l^{-1} at $\sigma_0 = 27.70$ and was therefore within the ACC. Station 12351 was on the boundary between ACC and subpolar waters, the UCDW signal being only just resolvable. Because of the dip in the isopleths, it appears that there were three SBs to ACC water, one boundary at station 12351 (63.4°S) (SB1), another between stations

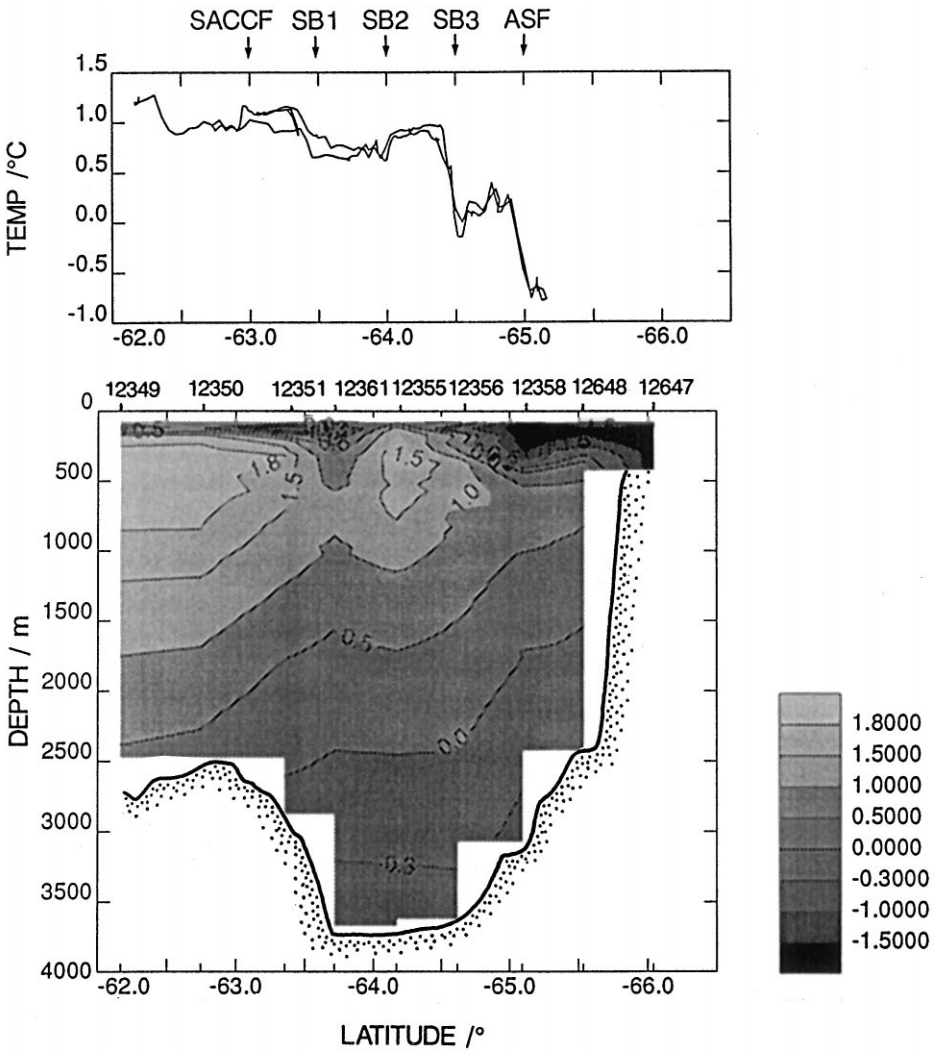

Fig. 2. The lower panel shows the CTD potential temperature section obtained in the Princess Elizabeth Trough during *RRS Discovery* cruises 200 and 207 in February 1993 and 1994. In the upper panel are the corresponding surface temperature data along track in 1993. Also shown are the CTD station locations and positions of the various fronts discussed in the text.

12361 and 12355 (63.7–64.2°S) (SB2) and a final one between 12355 and 12356 (64.2–64.6°S) (SB3). These boundaries also coincide with two surface temperature and salinity fronts shown in the TSG data (Figs. 2 and 3). Previously in published literature it was not known whether the SB had a surface signature, but this is the case here.

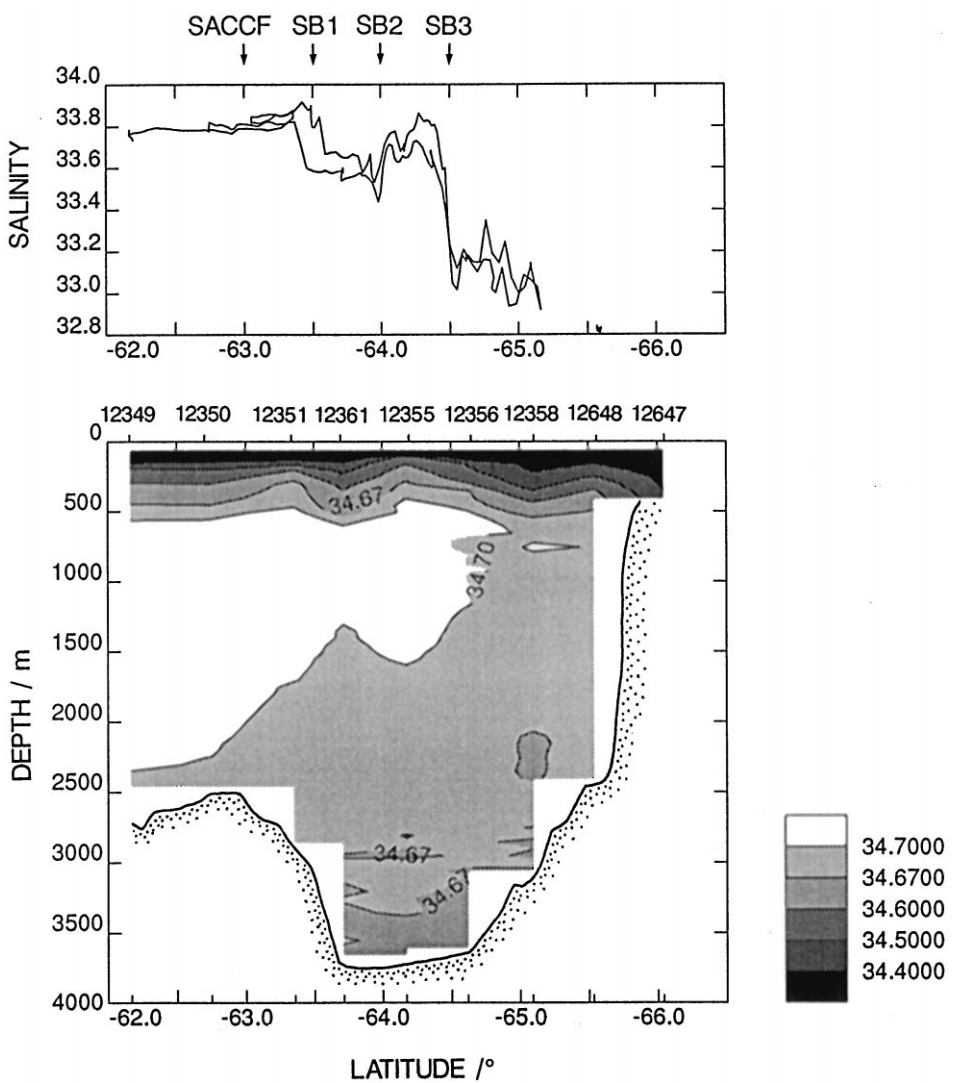

Fig. 3. As figure 2 but for salinity.

The SACCF can be identified at 63°S, more readily from temperature than salinity characteristics, only 0.5° of latitude (≈ 50 km) from SB1. Orsi et al. (1995) suggested that the SACCF is not associated with a surface temperature or salinity front, and this is confirmed by our TSG data. The isopleths sloping downward to the south in the CTD data show the northern edge of the ASF on the continental slope, associated with a surface temperature drop of about 1°C. This surface temperature front is consistent with observations in the Weddell Gyre (Fahrbach et al., 1992).

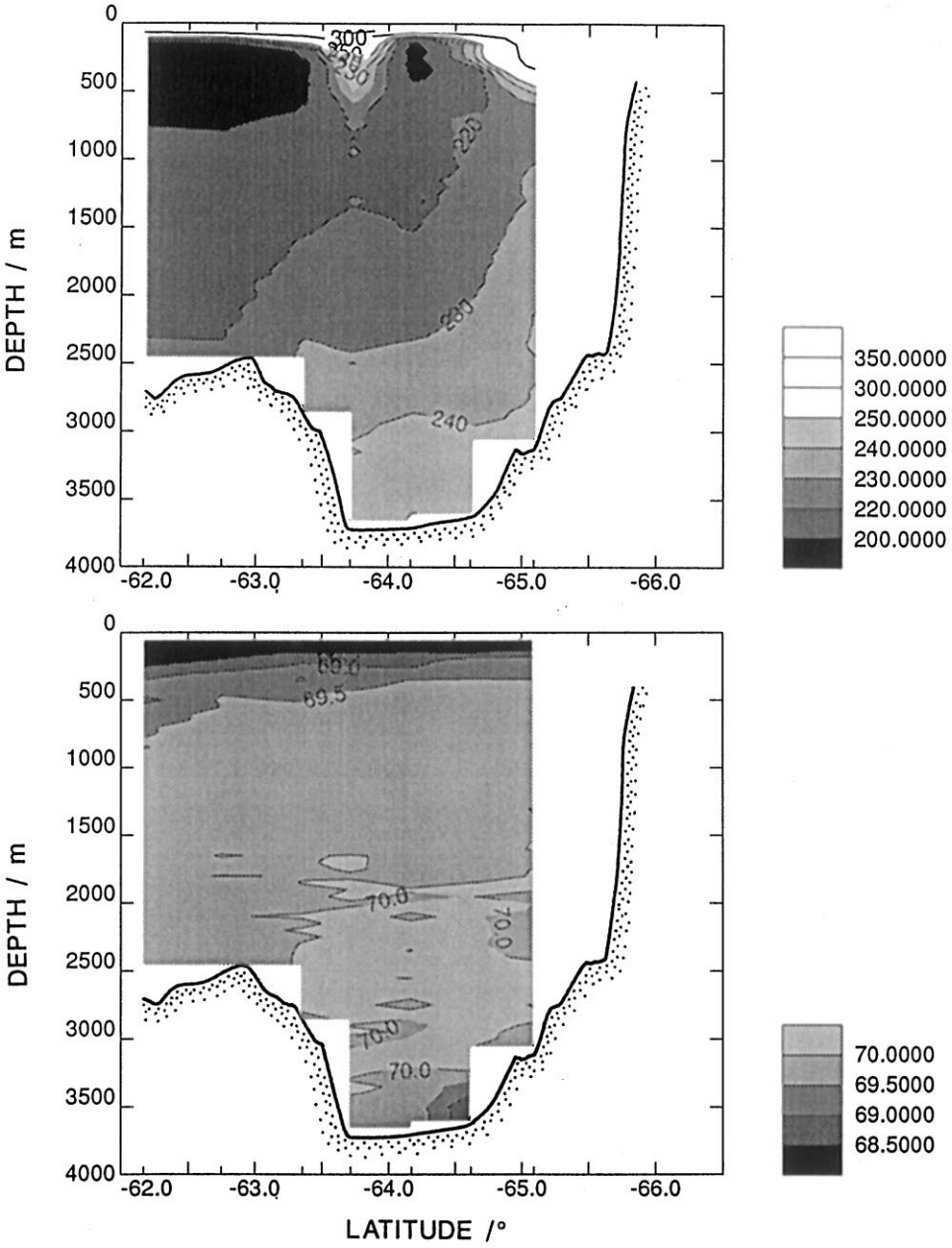

Fig. 4. Dissolved oxygen (micromoles per litre; upper panel) and potential transmittance (%; lower panel) sections obtained in the Princess Elizabeth Trough during *RRS Discovery* cruise 200 in February 1993.

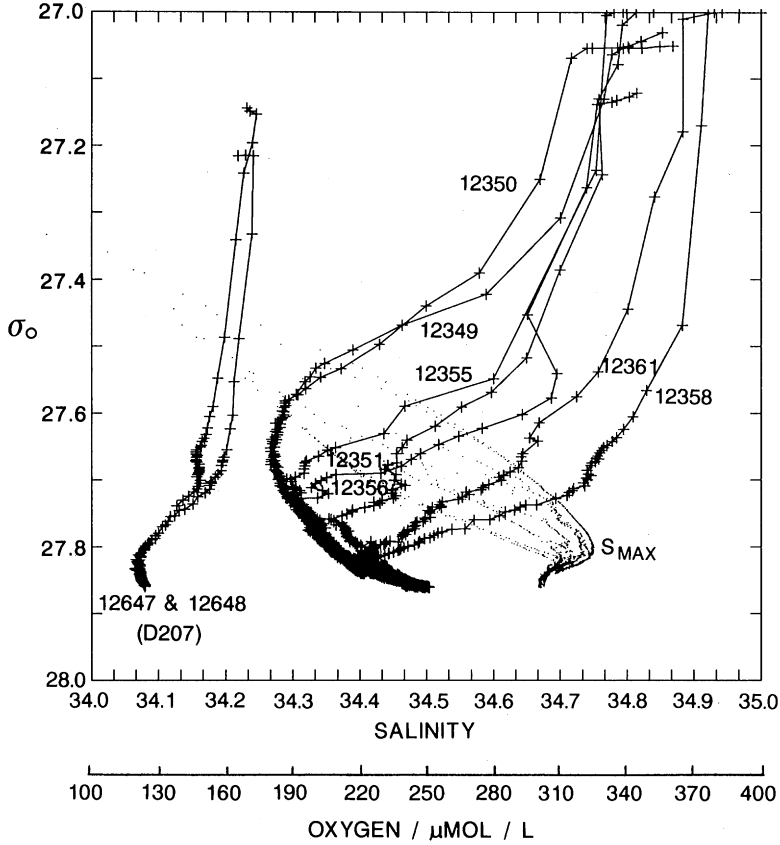

Fig. 5. Potential density anomaly (σ_0) against dissolved oxygen and salinity for all CTD stations occupied in the Princess Elizabeth Trough. Also shown is the salinity maximum S_{MAX} of Lower Circumpolar Deep Water (LCDW).

Geostrophic calculations relative to the deepest common level between stations (Fig. 6a) show at 63°S eastward flow of 10 cm s^{-1} at the surface and a transport of 10 Sv, corresponding to the SACCF and the SB1. Westward flow of 3 cm s^{-1} and 1.5 Sv occurs at 64°S (SB2), the velocity peaking in the subsurface layers at about 100 m depth. Eastward flow exists at 65°S of 10 cm s^{-1} and 12 Sv corresponding to SB3. Westward flow is observed on the upper continental slope associated with the ASF. The total transport across the section is 19 Sv eastward relative to the deepest common level. It is clear, however, that assuming zero velocity at the bottom is unlikely to be appropriate for many (if any) of these station pairs. Geostrophic calculations only yield velocity shear, and knowledge of the absolute velocity at any depth is required to estimate the barotropic component. We use the ADCP data to provide such a level of known motion.

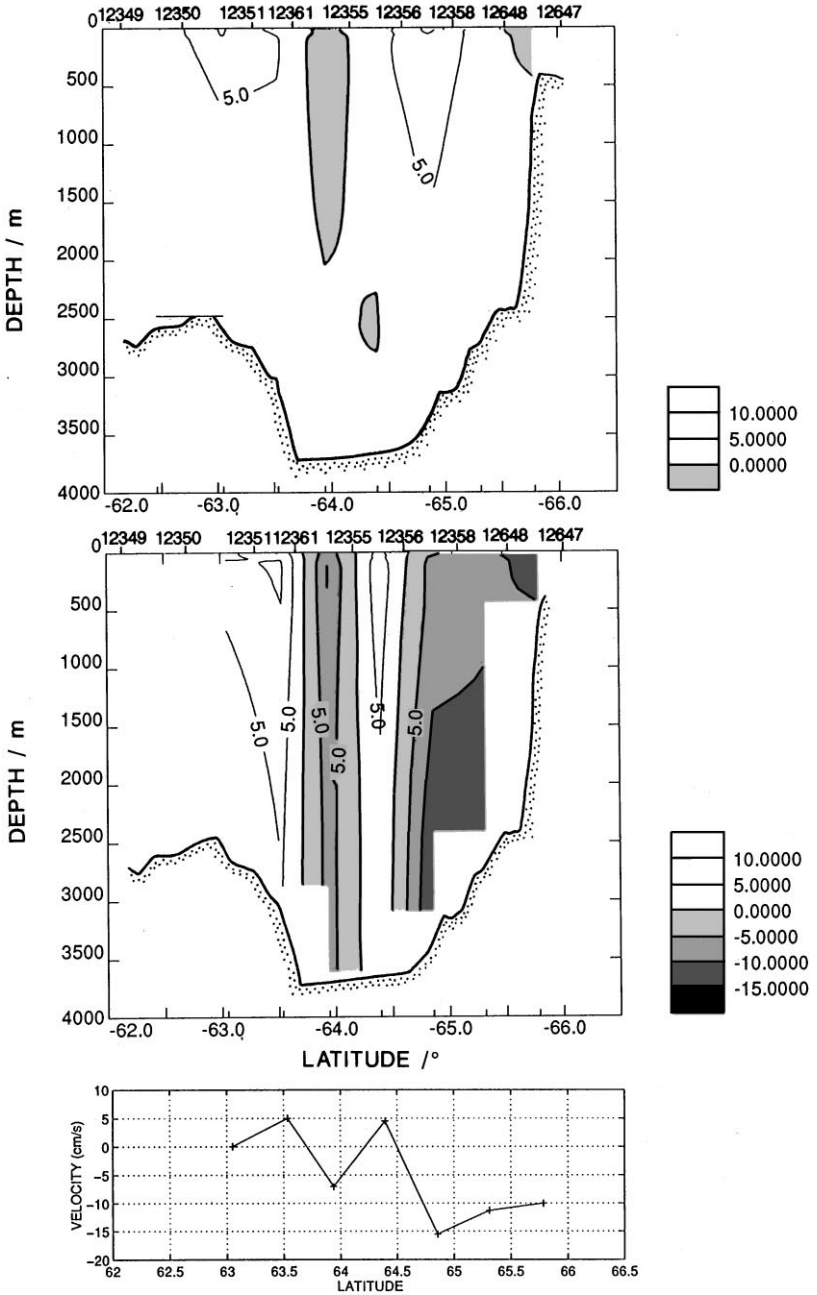

Fig. 6. Geostrophic velocity sections obtained in the Princess Elizabeth Trough during *RRS Discovery* cruises 200 and 207 in February 1993 (stations 12349–12361) and 1994 (stations 12648 and 12649). (a) Referenced to the deepest common level between stations. (b) Referenced to the near-surface ADCP velocities. (c) The bottom velocities applied to the geostrophic shears to derive the ADCP referenced velocities. Eastward speeds are positive and units are cm/s.

3.2. Acoustic doppler current profiler data

ADCP data within, and just to the west of, the PET at 200 m depth (below the mixed layer) are shown in Fig. 7. Within the PET in both years, the two crossings (from north to south and vice versa) show remarkably little temporal variability in the velocity field. We consider first the 1993 cruise (Fig. 7a); the 1994 cruise will be discussed in Section 3.4. The frontal jets associated with the SACCF (at 63°S) and SB1 (at 63.5°S) are not separately resolved despite the high horizontal resolution of the ADCP data. However at 63°S, peak velocities during both crossings are fairly constant at about 16 cm s^{-1}, whereas velocities at 63.4°S range from 18 to 40 cm s^{-1} at 200 m depth. Peak velocities associated with SB2 and SB3 are 20–28 cm s^{-1} and 20 cm s^{-1}, respectively, at 200 m. As far as we are aware these are the first observations of the absolute velocities associated with the SACCF and SB. Velocities in the ASF at 200 m depth peak at 28 cm s^{-1} (20 cm s^{-1} at the surface). To the west of the PET section velocities of up to 28 cm s^{-1} are seen in the ADCP data (Fig. 7a), possibly associated with the eastern boundary of the Prydz Bay gyre.

Underway ADCP data were selected for the steaming periods between adjacent CTD casts and averaged spatially to give the flow at the same position as the CTD geostrophic estimates. The cross-track shears of the geostrophic calculations relative to the deepest common level and of the ADCP data were compared. The effect of transient ageostrophic currents can be minimised by comparing shears below the

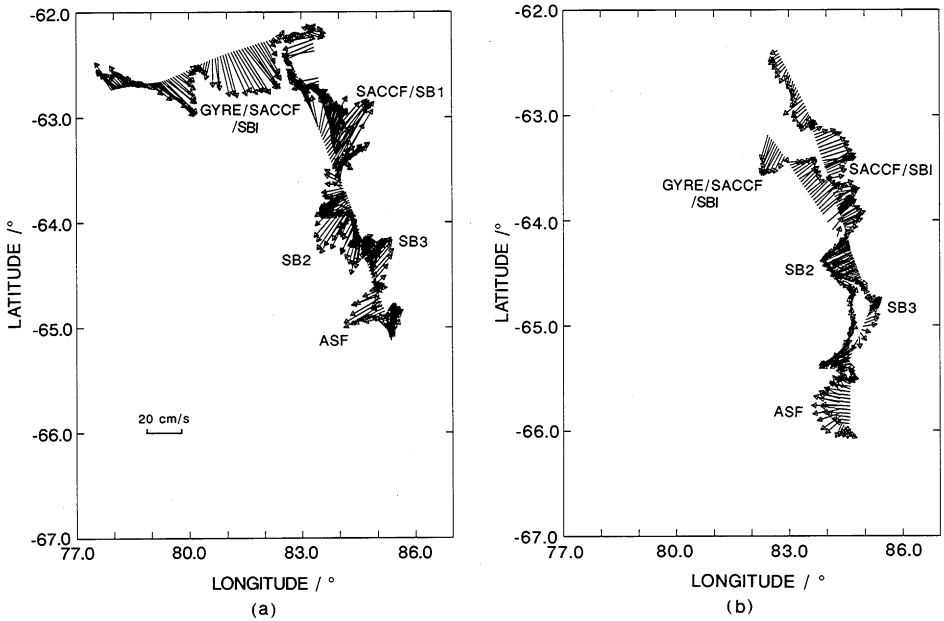

Fig. 7. ADCP data in the Princess Elizabeth Trough region at 200 m depth during (a) *RRS Discovery* cruise 200 in February 1993. (b) *RRS Discovery* cruise 207 in February 1994.

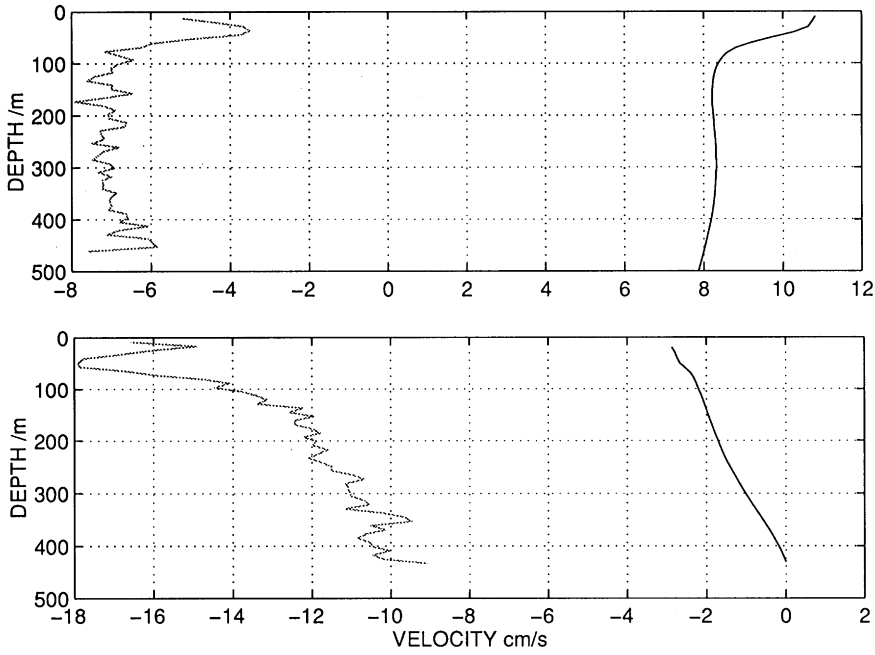

Fig. 8. Comparison of shears derived from CTD data relative to the deepest common level (smooth dark line) and underway ADCP data (jagged pale line). The upper panel is for stations 12356 and 12358, in the southern Princess Elizabeth Trough at 65°S. The lower panel is for stations 12647 and 12648, on the Antarctic continental slope at 66°S. Eastward flow is positive.

mixed layer (Saunders and King, 1995). If the shears are similar, the offset between the two profiles can be assumed to be the bottom velocity. The overall similarity between the ADCP current vectors on the north to south and south to north crossings is encouraging, as is the similarity between 1993 and 1994 (Fig. 7). This suggests that, below the wind driven layer, the balance of forces is predominantly geostrophic, and the flow structure varies little over the time scale of a week. Fig. 8 illustrates the comparison of the shears for two station pairs, the southernmost pairs from D200 (12356 and 12358) and D207 (12647 and 12648). Note that for depths between about 100 and 400 m, the shears are very similar. The offsets in both cases are, however, very large, 15 and 10 cm s^{-1}, respectively. These imply strong westward currents at the sea bed, which we shall see are associated with the flow of AABW between the Australian–Antarctic Basin and the Weddell–Enderby Basin, and with the ASF. In the case of the northernmost pair of stations from cruise D200 (12349 and 12350), there were gaps in the ADCP data, and the shears did not agree with those derived from geostrophy. This pair of stations has therefore been excluded from the analysis. It is to the north of the main core of the SACCF, and too shallow to be significant for AABW transport, so has no effect on our conclusions.

The bottom velocities that were derived from each offset are shown in Fig. 6c, with the new total geostrophic current (barotropic and baroclinic) shown in Fig. 6b. The uncertainty in the offset velocities is estimated as $\pm 2\ \mathrm{cm\,s^{-1}}$. In calculating total transports, account has been taken of the "bottom triangles" between adjacent CTD stations of differing depths. The meandering of the SB is now much more apparent; it is a bottom-reaching front with velocities in excess of $5\ \mathrm{cm\,s^{-1}}$. At the northern side of the PET, the combination of the SACCF and the SB has brought bottom water from the west (from the Weddell–Enderby Basin); its total transport through the whole water column between stations 12350, 12351 and 12361 (63 to 63.5°S) is 16 ($\pm$ 4) Sv to the east. As the SB meanders back to the west, its transport (between stations 12361 and 12355) is again 16 ($\pm$ 4) Sv. Its final eastward transport (between stations 12355 and 12356) is 11 ($\pm$ 4) Sv. The smaller value is probably because some bottom water is flowing westward in part of this section.

This meandering of the SB brings AABW from the Weddell–Enderby basin, which is dragged along in the meander by the strongly barotropic frontal velocities. Considering water with a potential temperature less than 0°C, transports are 2 ($\pm$ 1) Sv eastward, 5 ($\pm$ 1) Sv westward and 3 ($\pm$ 1) Sv eastward for the three crossings, from north to south, the average being 3 Sv.

In the southern PET, bottom water is transported to the west with velocities of order $15\ \mathrm{cm\,s^{-1}}$, and the volume of cold AABW is greater than in the northern PET, so resulting transports are an order of magnitude greater. The total westward transport of water less than 0°C in the southern PET at about 65°S is 20 ($\pm$ 3) Sv. The whole water column at this latitude is flowing to the west, as is the water on the continental slope and shelf; this is the flow associated with the ASF. The total full depth westward flow in the southern PET (between latitudes of about 65 and 66°S) is 45 ($\pm$ 5) Sv. This will be an underestimate, since some flow associated with the ASF will occur on the shelf, further south than our southernmost station. Notice that the velocity at the southernmost stations increases towards the surface; this is the flow that marks the ASF. Clearly nearly half of the 45 Sv is contributed by the bottom intensified flow of AABW from the Australian–Antarctic Basin into the Weddell–Enderby Basin.

The transports derived here are relatively large compared to previous estimates. We now consider evidence from current meters in the PET, to support the existence of large velocities in the deep and bottom waters.

3.3. Current meter data

In addition to the CTD and ADCP work, five moorings comprising 13 Aanderaa RCM5 current meters were deployed in a line across the PET (Fig. 1) at depths in excess of 2100 m. Meters at the bottom of two moorings failed to return speed data. More importantly, the majority of instruments suffered distinct compass problems similar in appearance to those reported by Hendry and Hartling (1978). The problem was manifest as a failure to produce data in significant segments of the compass with a concentration of values on the boundary of these segments. In the case of Hendry and Hartling the effect resulted from the magnetisation of the nickel coating by the

Table 1
Near-bottom speeds recorded at current meters in the Princess Elizabeth Trough during February 1993–February 1994. Units are cm s^{-1}

Latitude	63.1°S	63.5°S	63.9°S	64.4°S	64.8°S
Annual scalar mean	5.4	7.1	8.5	7.7	10.0
Maximum	25.0	27.6	31.4	31.7	39.5

compass magnets at high pressure when the permeability of the nickel was increased by the stresses induced in the pressure case. Our instruments were not nickel coated, and as yet we have no satisfactory explanation for the cause of the effect; therefore we present only speeds. The exception is the northernmost mooring (63.1°S), which shows little sign of the directional problem. This confirms an eastward flow in the northern PET associated with the bottom reaching SACCF. There the annual vector mean flow in water depth 2636 m at a height of 46 m above the bottom (too shallow to measure AABW) is 4.30 cm s^{-1} at an angle of 140°.

Table 1 presents speeds in the bottom water in the PET. These are averages of the one or two current meters within 500 m of the sea bed on each mooring. Tidal signals are of order 1 cm s^{-1}. For each mooring, we present the maximum and the annual mean for the period February 1993 to February 1994. Since CTD stations were almost exactly located midway between moorings, these speeds should correspond to the bottom flows deduced from the geostrophic currents referenced to the ADCP.

The trend is towards larger speeds in the southern PET, as was seen in the bottom velocities deduced using the ADCP (Fig. 6). Clearly even the value of 15.5 cm s^{-1} deduced for 64.8°S is not unusual given the annual mean speed of 10 cm s^{-1}, and the maximum recorded speed near 40 cm s^{-1}. Therefore our transports seem consistent with the annual values in this area.

Time series of current speeds in the southern PET (Fig. 9) show that the period of the cruise in February–March 1993 did not present anomalously large current speeds: the values are not dissimilar to the annual average. Currents are clearly variable throughout the year, although we have no evidence whether this is due to temporal variability in the volume transports, or spatial fluctuations in the locations of the various fronts moving past the moorings. There is no obvious seasonal or semi-annual signal in the current speeds, although a longer time series would be required to fully determine this.

3.4. Temporal variability of the flow structure

Limited data were also collected during 1994 (D207). The SACCF, SB and ASF had all moved about 0.5° further south during 1994. The XBT section and TSG data (Fig. 10) show the presence of warm water below about 100 m between 64.6°S and 65.0°S associated with the meander of ACC waters, though this is not as warm or well defined as during 1993 (Fig. 2). The location of the 1.8°C isotherm at depths greater than 500 m, and the 0°C isotherm at depths less than 150 m, position the SACCF at 63.3–63.9°S. Decreases in temperature mark the change from waters of the ACC to

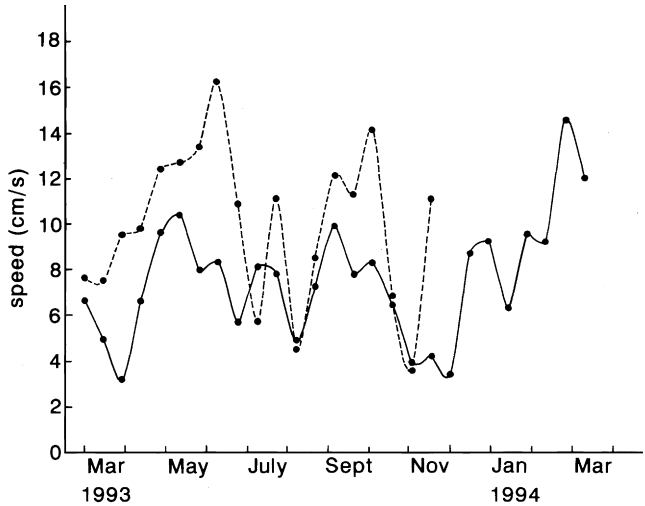

Fig. 9. Time series of speed at two current meter moorings in the southern Princess Elizabeth Trough from March 1993 to March 1994. The dashed line indicates the current meter at 64°49′S, 85°8′E (designated 9304-1), in 3367 m total water depth, 577 m above the sea bed. The solid line indicates the current meter at 64°23′S, 84°42′E (designated 9305-2), in 3654 m total water depth, 431 m above the sea bed. Speeds shown are 14 day averages.

those of the subpolar regime, indicating the SB. As during 1993, the SB is crossed three times, between 63.5°S and 64.0°S, at 64.6°S and at 65.0°S, suggesting meandering of the SB as observed during 1994, although the frontal jet associated with the SB is weaker. In 1993 the surface salinity is fresher (33.0) in the southern PET than in the northern PET. In 1994 the salinity remains relatively "high" (33.8) throughout the PET. This is likely to be caused by the lack of recent sea ice melt at the time of the cruise in 1994.

ADCP data (Fig. 7b) show peak velocities at 200 m depth in the SACCF of about 18 $cm\,s^{-1}$, comparable to the previous year. Velocities in the SB range from peaks of 14 $cm\,s^{-1}$ in SB1, 18 $cm\,s^{-1}$ in SB2 down to 8 $cm\,s^{-1}$ in SB3, generally lower than during 1993. The temperature minimum associated with the ASF is evident south of 65°S. Maximum velocities in the ASF are 28 $cm\,s^{-1}$ at 200 m depth, the same as the previous year. This consistency in flow speeds of the ASF contrasts with the degree of variability shown in the Weddell Sea (Fahrbach et al., 1992). This may be because the PET is less affected by tides. Peak velocities in the eastern boundary of the gyre are slightly less than in the previous year, at about 20 $cm\,s^{-1}$. As the section extends further south in 1994, the ASF can be seen to be double banded.

4. Discussion

The SACCF, identified from the criteria of Orsi et al. (1995), skirts the southern edge of the Kerguelen Plateau, passing through the PET at 63°S. This marks where

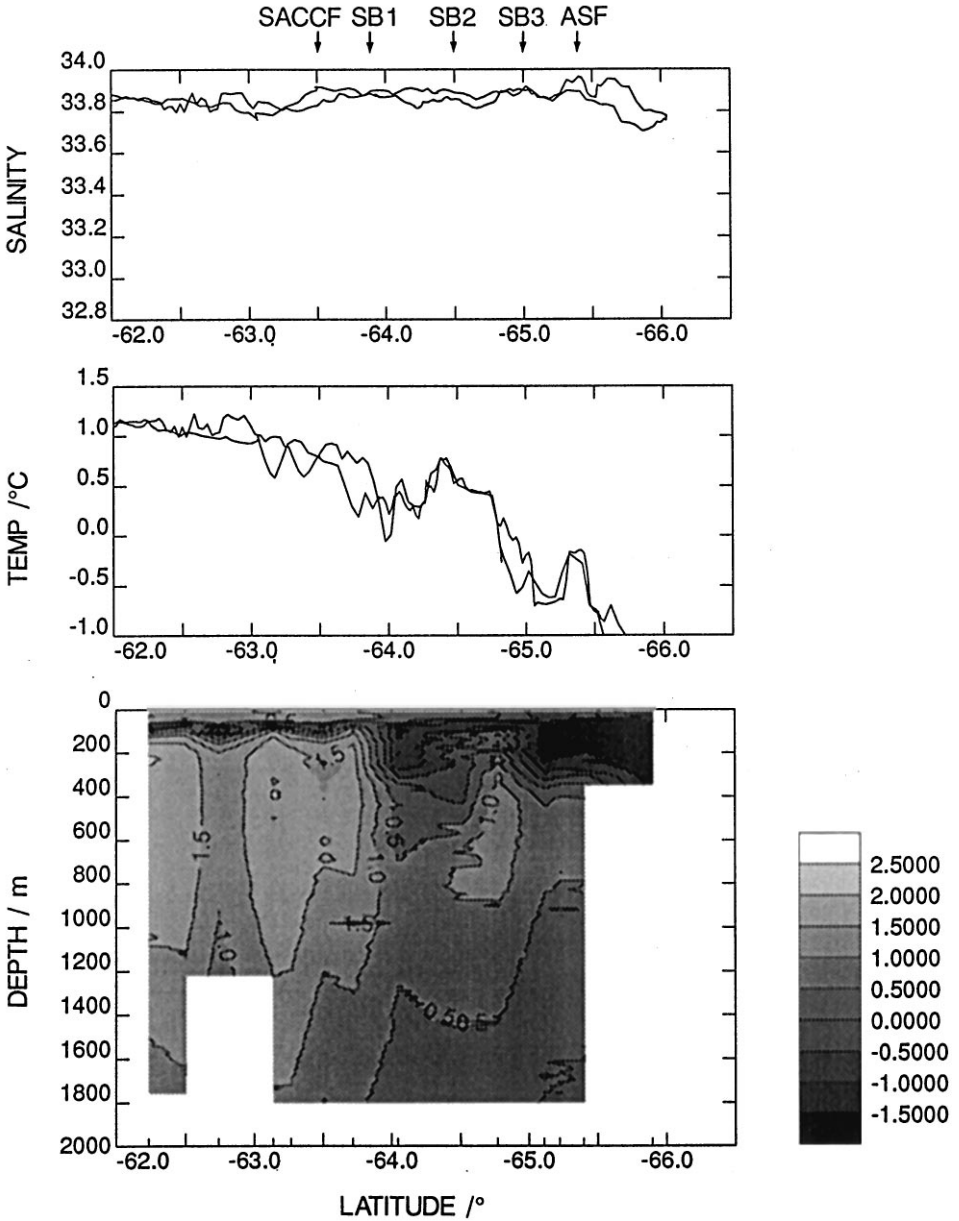

Fig. 10. XBT temperature section and surface temperature and salinity along track during *RRS Discovery* cruise 207 in February 1994. Also shown are the frontal positions.

the Antarctic Circumpolar Current extends over a very large range of latitudes. Analysis of historical data and numerical model output (Orsi et al., 1995; Sparrow et al., 1996) suggested the location of the front in the PET, which has been confirmed by the surveys described here. Peak velocities associated with the SACCF remain fairly consistent between the 1993 and 1994 cruises at 16–18 $cm\,s^{-1}$ at 200 m depth.

The SACCF remains in close proximity ($\sim 0.4^\circ$ of latitude or about 50 km) to the SB. In 1994 the SB was weaker than in 1993, with maximum temperatures in the ACC water about 0.5°C cooler at 500 m depth. There is a corresponding decrease in frontal velocities associated with the SB from peaks of 18–40 $cm\,s^{-1}$ at 200 m depth in 1993 to 8–18 $cm\,s^{-1}$ in 1994. During 1993 the SB is associated with strong surface thermohaline fronts, features which are much weaker the following year. As far as the authors are aware these are the first observations of the absolute velocities associated with the SACCF and SB.

This analysis has shown that the SB meanders in the PET, with a radius of about 50 km, flowing back to the west, and then east again in the deepest part of the PET (Fig. 1). If this meander had been observed on only one hydrographic section, the conclusion would be that it is an isolated eddy that has spun off from the ACC waters across the SB. We have seen that the same feature is apparent in three different years, 1992 (Speer and Forbes, 1994), 1993, and 1994. This meander appears to be due to the influence of a large gyre just to the west of the PET (spanning $\approx$ 15–20° of longitude) within the Prydz Bay area, centred on the Antarctic Divergence (Nunes Vaz and Lennon, 1996). A tongue of relatively warm and salty water of ACC origin is drawn cyclonically around the periphery of the gyre. The SB, identified by the southern limit of UCDW, follows the tongue of ACC water. This is also suggested in the Smith et al. (1984) map of surface temperature and salinity (their Fig. 6), and by the data presented by Nunes Vaz and Lennon (1996). These tongues of ACC waters are features commonly seen in the much larger Weddell and Ross Sea Gyres, for example by Orsi et al. (1993) in the Weddell Gyre and by Reid (1986) in the Ross Sea. Velocities in the eastern boundary of the Prydz Bay gyre are observed to peak at 20–28 $cm\,s^{-1}$ at 200 m depth, much faster than the 3 $cm\,s^{-1}$ surface speeds (which were only relative to 300 db) obtained by Smith et al. (1984). The primary meridional exchanges of heat and salt in the Antarctic are thought to be facilitated by these gyres and meanders of the ACC (the SACCF and SB) within the AD.

With the coarse CTD station spacing (about 50 km) of the surveys described here, it is not possible to separate the geostrophic transport of the SACCF from that associated with the SB. The total transport of the SACCF/SB is calculated to be 16 ($\pm$ 4) Sv. Read et al. (1995) obtained 15 Sv for the SACCF transport (referred to as the South Polar Front in their paper) relative to the deepest common level using full-depth stations at 85°W. At the Greenwich Meridian, the historical data analysed by Orsi et al. (1995) yielded 4.5 Sv for the SACCF and 3 Sv for the SB (both relative to 1000 m). Between 50°E and 70°E, again from analysis of historical data, Sparrow et al. (1996) deduced a combined flow for both frontal jets of 30 to 35 Sv. Our estimates, since they are the first to include the barotropic component of the flow, are likely to be the more accurate.

The absolute transport through the PET between 62 and 66°S is calculated to be 34 ($\pm$ 10) Sv westward. This is greatly different from the value of 19 Sv eastward calculated relative to the deepest common level between station pairs (or 12 Sv eastward relative to 2000 db). Speer and Forbes (1994) obtained a total transport across the PET of 7.9 Sv to the east, although they could make no estimate of the barotropic component of the flow which our analysis has shown to be important.

Velocities observed associated with the ASF are faster than the surface speeds of 4–15 cm s^{-1} inferred from ship drift (Jacobs, 1991) but comparable to the 10–20 cm s^{-1} surface current meter data from the Weddell Gyre (Fahrbach et al., 1992). Nunes Vaz and Lennon (1996) show that currents at 178 m depth from the shelf edge in Prydz Bay at 63°E are consistently strong and almost unidirectional westwards along the shelf break, with an annual average magnitude of 64 cm s^{-1}. The ASF exhibits strong seasonal and interannual variations in the Weddell Sea (Fahrbach et al., 1992) and might reasonably be expected to do so in the PET. The ASF shows minimal interannual variability in the velocity field, its associated frontal jet flowing westward with peak speeds of 28 cm s^{-1} at 200 m depth during both summer cruises. The ASF is associated with a minor surface thermohaline front.

In the northern PET, the meandering of the SB brings 3 ($\pm$ 1) Sv of AABW (defined by potential temperatures < 0°C) from the Weddell–Enderby basin, which is dragged along in the meander by the strongly barotropic frontal velocities. This eastward flow of AABW round the southern flank of the Kerguelen Plateau has also been suggested in the WOCE section I8S in 1994–95 (Hufford and McCartney, 1997). In the southern PET, the total westward transport of AABW is 20 ($\pm$ 3) Sv from the Australian–Antarctic Basin into the Weddell–Enderby Basin. The large transports and velocities in the southern PET are supported by several independent data sets, including the current meters and the high turbidity observed near the sea bed. Nevertheless, our estimate is likely to be an underestimate, for two reasons. Firstly, some westward AABW flow is likely between stations 12355 and 12356 and probably contributes to the lower transport deduced for the SB for that station pair. Secondly, there were few stations on the continental slope in the southern PET, so we may have missed a portion of cold AABW lying against the slope. Such westward transport of AABW from the Australian–Antarctic Basin was inferred by Mantisi et al. (1991) from chlorofluoromethane content in the southern PET, decreasing along the coast to the west. Speer and Forbes (1994) estimated that 3.7 Sv (using a varying reference level) of "deep and bottom waters" flows westward through the PET, but this will be a total of the two opposing flows. It is therefore not surprising that their values are so much lower; theirs is a net transport, and was also derived without the knowledge of the barotropic nature of these frontal jets.

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